Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and the Labor Market (Listas 3–4)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Consumption and saving

(Kurlat ch. 6 • 25 points)

Question 1. (10 points) Objective items. A young worker in São Paulo lives two periods, with $u(c_1) + \beta u(c_2)$, $u = \ln c$, $\beta = 0.8$, income $y_1 = 60$, $y_2 = 150$, and borrows or lends freely at $r = 0.25$ (illustrative; a period is a decade).

a) (3 points) The worker's optimal saving $a = y_1 - c_1$ is:

A. -30

B. $+40$

C. -40

D. -20

b) (3 points) Replace log utility by $u(c) = c^{1-\sigma}/(1-\sigma)$ with $\sigma = 2$, keeping β, r and incomes. Then c_1 equals:

A. 120.0

B. 100.0

C. 90.0

D. 108.3

c) (2 points, T or F) For the worker in (a), a tax cut of 10 in period 1 financed by a tax increase of 12.5 in period 2 raises c_1.

d) (2 points, T or F) If the worker faces the borrowing limit $a \geq -10$, the same tax change raises c_1 by exactly 10.

Question 2. (15 points) Discursive. A worker earns y in period 1 and nothing in period 2 (retirement). Preferences $u(c_1) + \beta u(c_2)$ with $u(c) = c^{1-\sigma}/(1-\sigma)$ ($\ln c$ if $\sigma = 1$); the worker saves at rate r.

a) (5 points) Derive c_1, c_2 and a in closed form. Show how the sign of $\partial a/\partial r$ depends on σ, and explain it with income and substitution effects.

Solution:

b) (5 points) Let $\sigma = 1$, $\beta = 0.8$, $r = 0.25$, $y = 150$. The government taxes interest income at $\tau = 0.4$, so $c_2 = (1 + (1-\tau)r)a + T$, and returns the revenue as a lump-sum T in period 2 (the worker takes T as given). Compute c_1, c_2, a and T with and without the tax. Show that the tax changes c_2/c_1 while an equal-revenue lump-sum tax would not, and measure the welfare cost as the percentage cut in consumption in both periods that would give the untaxed worker the same utility.

Solution:

c) (5 points) Return to the worker of Question 1(a) ($y_1 = 60$, $y_2 = 150$, log, $\beta = 0.8$, $r = 0.25$). Compare the response of c_1 to (i) a one-off transfer of 10 in period 1 (like the 2020 *auxílio emergencial*) and (ii) a permanent rise of 10 in both periods. Redo both for a worker facing $a \geq -10$. What does this say about the permanent income hypothesis and about the Keynesian consumption function?

Solution:

Block 2 — Labor supply and unemployment *(Kurlat ch. 7 • 25 points)*

Question 3. (10 points) Objective items. A PNAD Contínua respondent has $u = \ln c + \theta \ln l$, $\theta = 1$, one unit of time ($h = 1 - l$), real wage $w = 20$ and non-labor income $\pi = 4$.

a) (3 points) Hours worked are:

A. 0.6

B. 0.4

C. 0.5

D. 0.2

b) (3 points) The reservation wage (below which she does not work) is:

A. 2

B. 8

C. 20

D. 4

c) (2 points, T or F) For this respondent, a 10% rise in the after-tax wage leaves hours unchanged, because with log utility income and substitution effects cancel.

d) (2 points, T or F) In the DMP model with $m = \mu V^{\alpha} U^{1-\alpha}$, a rise in matching efficiency μ shifts the Beveridge curve toward the origin: fewer vacancies are needed at each unemployment rate.

Question 4. (15 points) Discursive. Brazil's labor tax wedge. A household has $U = \ln c + b\,\dfrac{l^{1-\gamma}}{1-\gamma}$ ($b, \gamma > 0$), one unit of time, wage w, labor-income tax τ and a lump-sum transfer T: $c = (1-\tau)w(1-l) + T$.

a) (5 points) Derive the labor supply condition. Show that when $T = 0$ hours depend on neither w nor τ, for any γ. Show that when $T > 0$ hours rise with $(1-\tau)w$.

Solution:

b) (5 points) Now the budget balances: $T = \tau w(1 - l)$ in equilibrium (each household still takes T as given). With $\gamma = 1$, $b = 1$, find hours as a function of τ, compute them for $\tau = 0$ and $\tau = 0.5$ and $dh/d\tau$ at $\tau = 0$. Why is the effect larger than in (a) with $T = 0$? Relate to Prescott's comparison of hours in Europe and the US.

Solution:

c) (5 points) DMP model: separation rate $s = 0.03$ per month, $m = \mu V^{0.5} U^{0.5}$, $\mu = 0.6$, tightness $\theta = V/U = 0.25$. Compute f, q, u^* and v^*. After the pandemic, mismatch lowers μ to 0.45. Compute u^* at the same θ and the vacancy rate needed to keep u at its old value. Explain the shift versus a movement along the Beveridge curve, and why a firm posting one more vacancy creates an externality.

Solution: